Lab6: Find All Simple Paths on Random Adjacency Matrix

You will write a Python program that discovers **all possible simple paths** (no repeated nodes) between two nodes in a randomly generated directed graph.

Your program must include:

1. Random Number of Nodes

- The program must generate a random number of nodes:
  - N should be between **3 and 5**, inclusive.
  - The user does **NOT** choose this number.
  - The program prints the randomly chosen node count.

2. Random Adjacency Matrix

- Generate an N×N adjacency matrix.
- For each pair of nodes (i, j):
  - If i == j, value is 0 (no self-loop).
  - Otherwise, randomly choose 0 or 1.
- Interpret:
  - 1 → There is a directed edge from node i to node j
  - 0 → No edge

3. User Input With Exception Handling

The user will enter:

1. **Start node index**
2. **End node index**

Both inputs must satisfy:
- Must be **integers**
- Must be between **1 and N**
- If user enters:
  - A number **less than 1**
  - Or **greater than N**
  - Or **non-integer**

…then you **must raise an exception** and display a meaningful error.

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4. Find All Simple Paths

- Collect all **simple paths** from Start → End.
- A simple path means:
  No node may appear more than once in a path.

5. Output

- Print all discovered paths using 1-based indexing.
- If no path exists, print: No paths exist between X and Y.

Input	**Output**
Randomly selected number of nodes: 5 Adjacency Matrix: [0, 1, 0, 1, 0] [0, 0, 1, 0, 0] [0, 0, 0, 1, 1] [0, 0, 0, 0, 1] [0, 0, 0, 0, 0] Enter start node (1..5): 1 Enter end node (1..5): 5	1 -> 4 -> 5 1 -> 2 -> 3 -> 5 1 -> 2 -> 3 -> 4 -> 5
Randomly selected number of nodes: 4 Adjacency Matrix: [0, 1, 0, 0] [0, 0, 1, 1] [0, 0, 0, 1] [0, 0, 0, 0] Enter start node (1..4): 1 Enter end node (1..4): 4	1 → 2 → 3 → 4 1 → 2 → 4